



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

DIGITAL LOGIC
SECR1013
GROUP 9

Project Title	Lift Electronic Controller System
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1. ACKNOWLEDGEMENT

On this project, we worked quite hard. That being said, it would not have been achievable without the kind support of a lot of people. We want to express our sincere gratitude to everyone. For their inspiration and support in helping us finish this project, we would like to express our gratitude to our parents and other family members.

In addition, we would like to thank and honour everyone who helped us in any way with developing the case report. Thanks to our wonderful instructor, Dr. Nurfazrina, for her assistance and guidance, our case study was successful. We could not have finished our project by the deadline without her assistance.

We really appreciate all of our fellow friends who encouraged and supported us in completing this fantastic endeavour for Digital Logic (SECR1013).

2. BACKGROUND

For the purpose of this project, we created a card holder detection as a password to verify that the card matches the input, which is the level that the user pressed. In order to help the user know that they have reached the desired level, we additionally inserted the light bulb, as it will display the floor number when the lift has arrived at the desired level. Furthermore, we added a "HELLO" text welcome to the elevator. The full circuit is drawn using deeds and includes combinational circuit and sequential circuit.

Combinational circuit components include:

- 3-bit comparator
- Basic gates
- Switches
- LEDs
- Seven segment display

Sequential circuit components include:

- 3 bit up Counter
- Clock disabler

Enhanced features:

- 4 digit password
- Level indicator
- Text greetings

3. PROBLEM STATEMENT

The existing hotel elevator system lacks an optimal user experience and effective security features. The challenge is to create a better system so that guests can enter the floor they want to go to, use their room cards to access the elevator, and travel between floors safely and effectively. By taking care of these requirements, the hotel's elevator system will function more safely and more easily. In addition, to make our lift more user-friendly, we have also added some additional features, such as a welcoming LED sign.

4. SUGGESTED SOLUTION

Lifts are a major point of vulnerability in various buildings including our hotel, putting our guests and belongings at danger of vandalism, theft, or even bodily harm. To let the users experience an optimal satisfaction and security in our hotel, we added a lift card access system to our lifts. When lift card access system is implemented, the lifts will only open for an authorised person who presents his or her card, allowing us to closely manage access to your building's levels.

TRUTH TABLE

Input	Present State			Next State			D FF		
X	Q2	Q1	Q0	Q2+	Q1+	Q0+	D2	D1	D0
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	1	1	0	1	1
0	0	1	1	1	0	0	1	0	0
0	1	0	0	1	0	1	1	0	1
0	1	0	1	1	1	0	1	1	0
0	1	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	1
1	0	1	1	0	1	0	0	1	0
1	1	0	0	0	1	1	0	1	1
1	1	0	1	1	0	0	1	0	0
1	1	1	0	1	0	1	1	0	1
1	1	1	1	1	1	0	1	1	0

K-MAP

D2 K-MAP

Q1Q0 \ XQ2	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	0	1	1	1
10	0	0	0	0

$$D2 = (X'Q2) + (X'Q1Q0) + (Q2Q0) + (Q2Q1)$$

D1 K-MAP

Q1Q0 \ XQ2	00	01	11	10
00	0	1	0	1
01	0	1	1	1
11	1	0	1	0
10	0	0	1	0

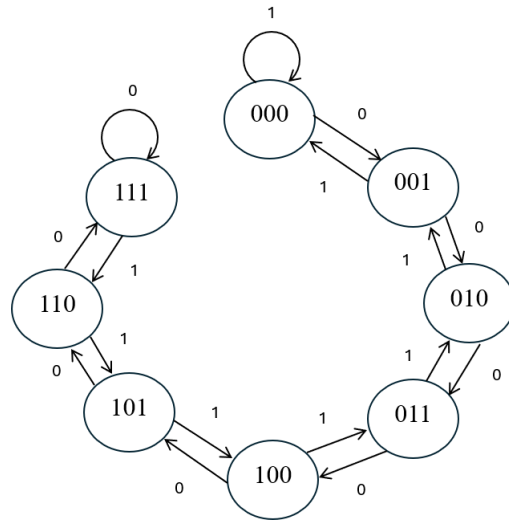
$$D1 = (X'Q1'Q0) + (X'Q2Q1) + (XQ1Q0) + (X'Q1Q0') + (XQ2Q1'Q0')$$

D0 K-MAP

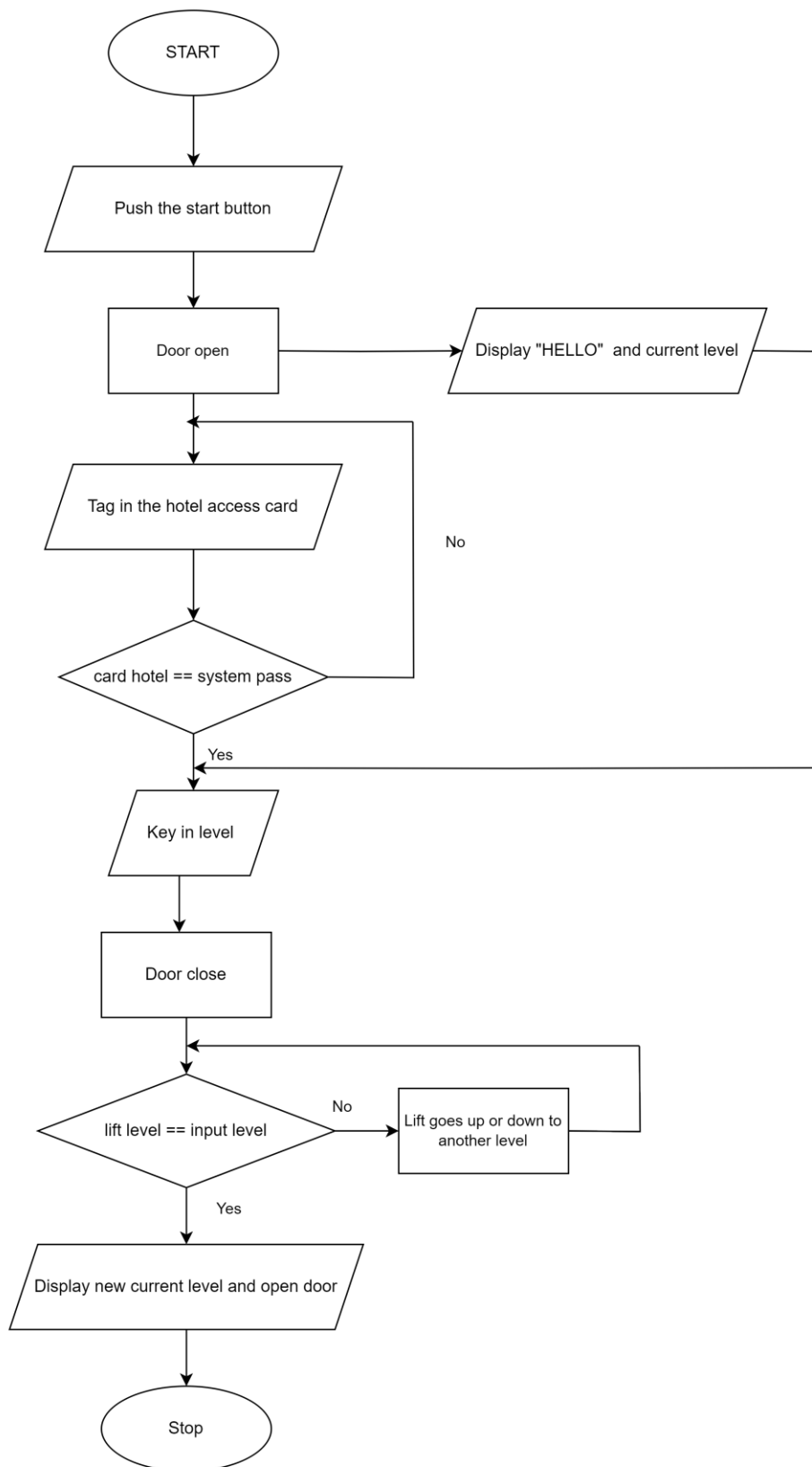
Q1Q0 \ XQ2	00	01	11	10
00	1	0	0	1
01	1	0	1	1
11	1	0	0	1
10	0	0	0	1

$$D0 = (X'Q0') + (Q1Q0') + (Q2Q0') + (X'Q2Q1)$$

STATE DIAGRAM



BLOCK DIAGRAM



5. REQUIREMENTS

This project requires the integration between combinational circuit and also sequential circuit. We are required to include combinational circuit in our project and these are the components used :-

- 3-bit comparator

The user's input will be compared with the binary value that we specified as our password using this 3-bit comparator.

- 7 segment display

This will display as a floor number that the user desired to go. We use 7 segment because we have 7 levels ; ground floor,1,2,3,4,5,6,7. display because there are & We used to display these numbers by using switches.

- 3-bit Decoder

It uses to translate or decode coded information from one format into another. Therefore, a binary decoder transforms 3 binary input signals into an equivalent code using 2^3 outputs.

- 3-bit D counter

3 bit D counter flip flop is a modified version of Set-Reset flip-flop with the addition of an inverter to prevent the S and R inputs from being at the same logic level. Here are the D excitation table, as it help us to do the Truth Table :-

Present state	Next state	FF state
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

- Basic gates

The basic gates that are used in this circuits are AND, OR, NOT, NAND, and XNOR. The basic gate is the fundamental of our system.

- Switches

The switches are function as a start button to start the operation.

- LEDs

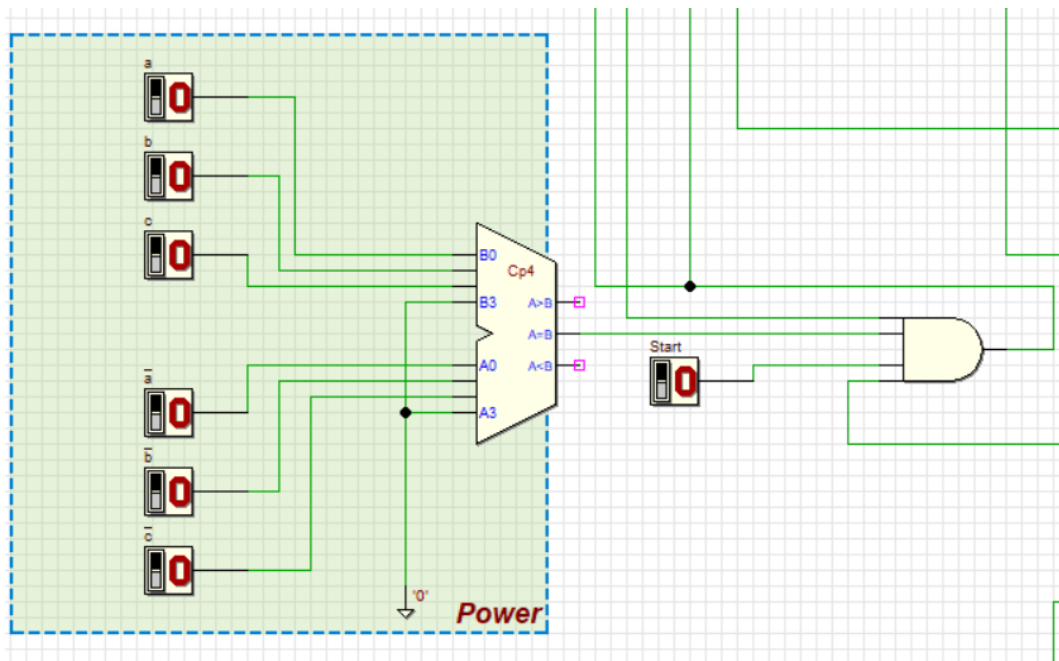
The LEDs are used to design as the text for “Hello”, indicator for door open and close, and display the desired floor from ground floor (g) to level 7.

- Clock disabler

Causes the flip flop either to change or to retain its output signal based upon the value of the input signals at the transition. In our system, the output change on the rising edge of the clock.

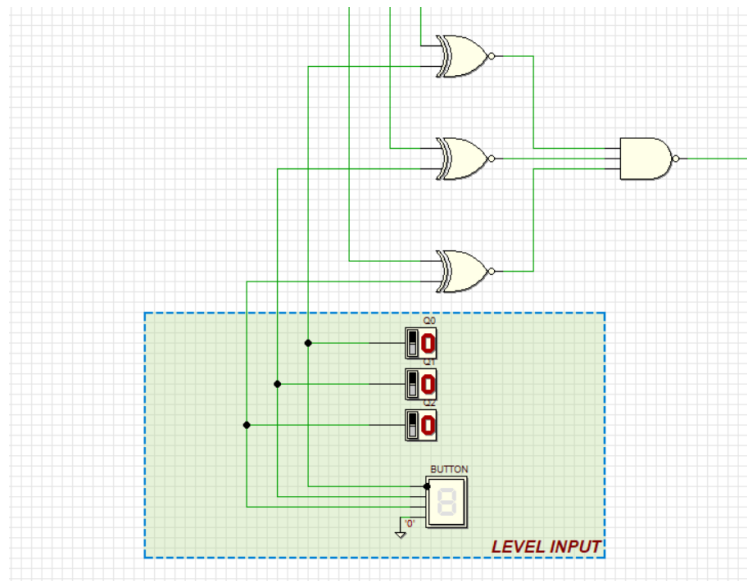
6. SYSTEM IMPLEMENTATION

1. 4 - Bit Comparator (MSI CHIP) (for password)



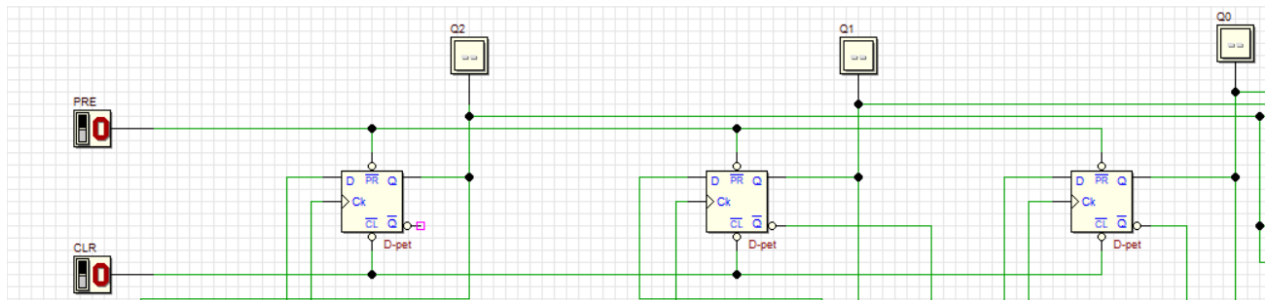
This 4 bit comparator is used to compare the input by the user with the binary value that we set as our password. After the values are compared among each other, the lift will operate if the password entered is right and it will not operate and remain on the same level if the password is wrong. We connected B3 and A3 to ground since we only need 3 bit input.

2. 3 – bit Comparator (to compare level)



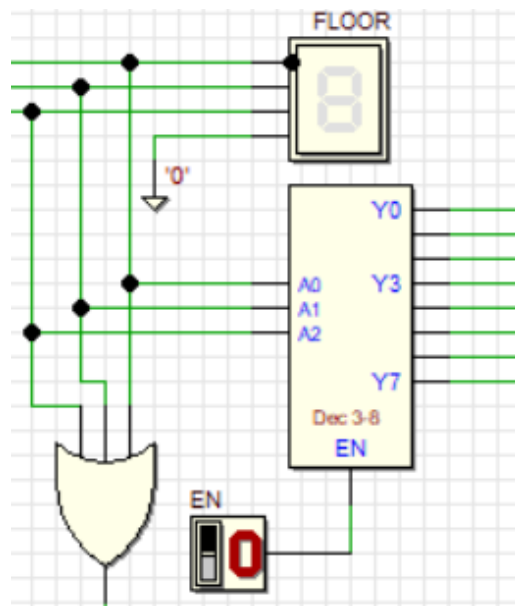
The XNOR gates are used as a basic comparator specifically to determine the equality which output 1 indicates the equality and 0 indicates the inequality. For this circuit, this 3 – bit comparator is used to compare the levels chosen by the user. The inputs for this 3 – bit comparator are Q2, Q1 Q0 and also the chosen level.

3. 3 – bit D Counter



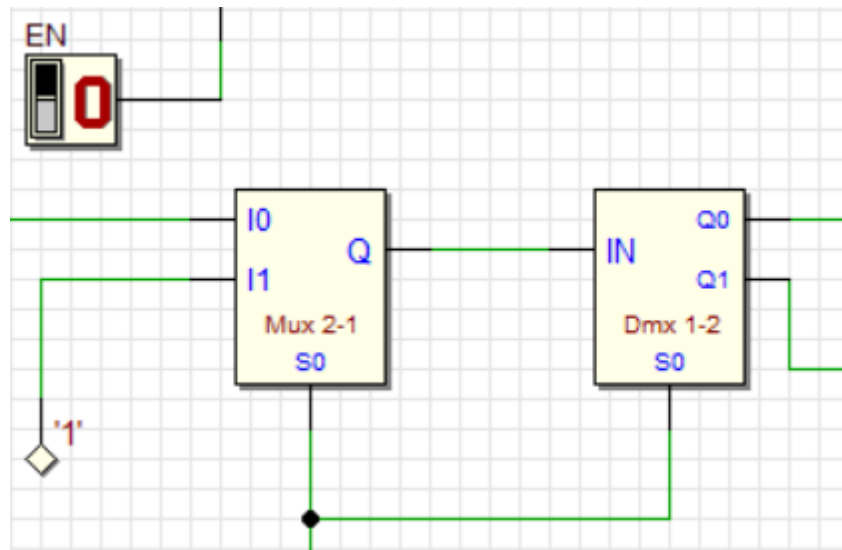
We use 3-bit bidirectional saturated D Counter to indicate how many levels the user can choose. There are only 7 floors. The level will increase or decrease each time the clock is pressed. The counter will stop if either the password is incorrect, or the power switch or clock enabler have low input.

4. 3 – bit Decoder



To display the floor where the lift is located, we utilised a 3 to 8 decoder. This decoder has eight outputs that are available. If the input is 101, output number 5 will be selected. Same applies to other binary inputs, such as 110, 100, and so forth. In order to show the output in decimal format, we linked the decoder to an LED.

5. Multiplexer and Demultiplexer



The multiplexer 1 – of - 2 installed allows digital information from several sources to be routed onto a single line for transmission over that line to a same destination. The basic multiplexer has several data-input lines and only a single output line which is Many-to-One. Meanwhile, the demultiplexer works vice versa. It takes digital information from one line and distributes it to a given number of output lines. For our circuit, the multiplexer 1 – of – 2 is able to route the floor desired by the user and connect it to the demultiplexer while the demultiplexer distributes it to the open and close LED arrows that have been designated.

7. CONCLUSION AND REFLECTION

a) Summarizes the whole project. Write your reflections.

- **Yasmin Batrisyia**

By completing this project, it has improved in terms of my knowledge in digital logic, and it has given me the opportunity to apply the knowledge that we have learned in class about designing a system and express our creativity and innovative skills in designing a good and user-friendly lift controller system. I would like to thank my lecturer, Dr. Nurfazrina for her assistance and help in completing this project. I strongly believe that this project is really important for us as we could gain the knowledge, skills, and hands-on experience in building a complex system until it is successfully operating as what we planned. This is because it will greatly benefit us in the future as this is a preparation for us to the real world of work.

- **Syarifah Dania**

This project optimizes our knowledge and skills in using Deeds Software, we were also able to enhance our problem solving skills because this project requires us to implement the basics of combinational and sequential logic circuit that we learnt in class to a real-life situation and problem. Also, as this project requires a team effort and work, it helps us to improve our communication skills and also increase the efficiency of our project development as we all worked collaboratively to divide responsibilities, with team members focusing on different aspects such as the logic circuit design, video demonstration and the end report . To sum it all up, I think this project made us able to deeply understand what we have learnt in class for this semester and I am very grateful to be able to finish this project in time with the immense help and assistance from my groupmates and my digital logic lecturer, Dr. Nurfazrina.

- **Nurul Asyikin**

The concept behind this project is that students will have to use their creative skills and knowledge to construct a lift system. By doing this project, it encourages us, students to develop critical thinking abilities and create innovative solutions for real-world problems by conducting research and having a thorough comprehension of all the information that must be used. In summary, this project serves as a platform for me to thoroughly explore components, their functions, and the real-world implementation of these components.

- **Ain Nurnabila**

By completing this project, I embarked on a new interest which is by completing the Lift Electronic Controller System Circuit. I managed to apply and also improved my knowledge on both combinational circuit and sequential circuit including complex circuit. This project helped me in enhancing my creativity and also high order thinking skills in order to design this project. In order to finish this project, the exposure that I received really helped me to prepare myself for real life-work situations. I am beyond grateful to Dr. Nurfazrina for giving me this kind of opportunity.

b) Describe your achievements, strength and weaknesses.

- **Syarifah Dania**

This project allowed me to understand what I had learned in the subject of digital logic on a deeper level. In addition to fulfilling my responsibilities as a group member, I was able to help my fellow teammates finish this project. Our work was assisted by my group members and me exchanging ideas and knowledge. I asked for the assistance of my friends because I am really bad at building a circuit that works correctly. With their help, I was able to understand every function of the circuit, which improved my comprehension of the parts and their functions.

- **Yasmin Batrisyia**

My achievements throughout this project are I get to strengthen my knowledge of various components that could give benefit to the system such as Decoder, Comparator and all the use of basic logic gates. This is because, to create the nest system, I have to really study deeply about every component in detail and elevated the knowledge that I gain to the Deeds software. My strengths are I'm hard-working person. So, when I got the task about the project, I quickly have a discussion with my group members to complete this project and create the best system as we can. About my weaknesses are I'm lack of knowledge in digital logic. Therefore, it takes a long time for me to complete the project as I must study properly form various of reference such as books, internet and discussing with my group members to really understand the use for every component in building the system. Due, to my hard work and dedication, me and my friends managed to successfully do our project in flying colours.

- **Nurul Asyikin**

My main achievement in making this project a huge success was being able to comprehend thoroughly each component and function as well as how they interacted with one another. By researching the system thoroughly and welcoming feedback from others, I was able to improve my strengths in interpersonal communication, which will be essential for future teamwork success. From this project, I realize that my weakness is that I lack the creativity and expertise to fully utilize the design; as a result, I count on teammates for guidance and encouragement. I learned from this assignment that collaboration is essential to producing an excellent outcome.

- **Ain Nurnabila**

Things that I have achieved in the process of completing this project is I got to enlarge my knowledge on the components used in everyday real-life situation circuits. Without knowing deeply about the components such as decoder, comparator, etc, I would not be able to design a Lift Electronic Control System Circuit. Throughout the process, I got to be aware of my weakness. I learned a lot from my groupmates about the usage of the components I rarely used before in deeds simulation. On the other hand, I also managed to acknowledge my strength which I think I did well in expanding the creativeness of the circuit. Not to forget, my groupmates also spent their creativity in designing this complex circuit. In my opinion, it was a good beginning for first-timers like us.

c) **Propose future work to improve your design**

- **Syarifah Dania**

I think that in order to improve this design, I need to perform a lot of study and fully understand each part and its functions. With my deep understanding, I can create an ideal circuit and get the desired result. As there are a lot of components in this project, I'm thinking about seeking up some examples online to help me stay organised. Some of the expert's examples or works could help me come up with some ideas and boost the content of my work more significantly.

- **Yasmin Batrisyia**

In my opinion to improve my designing skills, I have to do a lot of revision about all the components that I have learned in this course, and study them deeply on how it is function in order to strengthen my basics. I plan to enhance my knowledge in designing the circuit by learning other components too and also watch how other people doing through YouTube, Google and so on. In a nutshell, I have to really increase my understanding in order to generate a quality and creative system.

- **Nurul Asyikin**

Regarding how I intend to advance my designing abilities, I will begin by thoroughly reviewing every component I learned during this course, studying them in depth, and becoming familiar with all of their uses and functions in order to strengthen my foundation. I also want to expand my skills by studying other components and observing how other people have designed their systems on different websites, like Google, YouTube, and so forth.

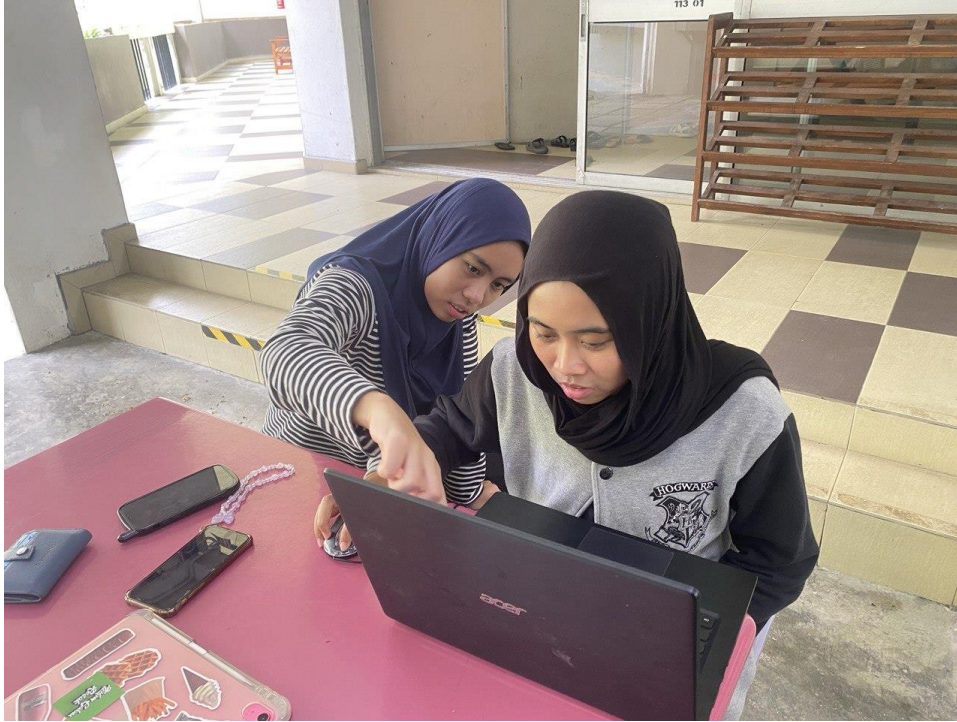
- **Ain Nurnabila**

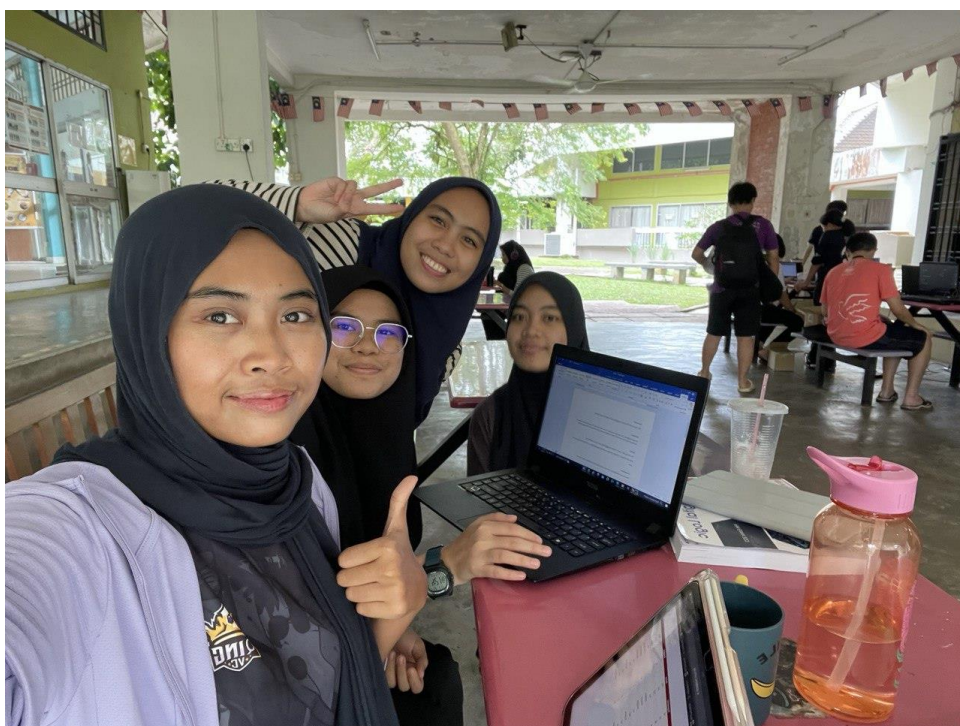
In my opinion, in order to improve my work progress in the future, I have to study about the components used beforehand. This step will prevent myself from being clueless at the first stage of doing the project. Moreover, I also have to gain and collect information from various sources in order to widen my sight of the project. I think by doing this, I will get to improve my skills in designing a circuit henceforward.

8. REFERENCES

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APPENDICES





TASK DISTRIBUTION

Nurul Asyikin

→ Report

- Problem statement
- Suggested solution
- Conclusion & Reflection

→ Video

- Sound editing
- Video editing

Syarifah Dania

→ Report

- Suggested Solution
- Background
- Conclusion & Reflection

→ Video

- Project demonstration voice over
- Conclusion voice over

Yasmin Batrisyia

→ Report

- Background
- Requirements
- Conclusion & Reflection

→ Video

- Sound editing
- Video editing

Ain Nurnabila

→ Report

- Requirements
- System Implementation

- Conclusion & Reflection
- Appendices

→ Video

- Problem statement voice over
- Solution voice over
- Requirements voice over

Deeds Simulation

→ Everyone